Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Development of High-Efficiency Soft Magnetic Components based on Amorphous Alloy Microwires for Next-Generation Power Electronics			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility (), Next-gen Nuclear Power ( ), Advanced Bio (), Aerospace/Marine (), Hydrogen ( ), Cybersecurity ( ), AI (), Next-gen Communication (), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical – Materials		Electrical – Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Metal Materials	Functional Materials	Electrical and Electronic Components	Magnetic Material Components
Concept of Proposed Technology	The project proposes the integration of amorphous metal microwires, known for their excellent magnetic permeability and superior high-frequency performance, into soft magnetic composite (SMC) components with high saturation polarization and ultra-low coercivity. Conventional approaches derived from amorphous ribbons are typically limited to packing densities of approximately 80%. In contrast, the compaction of amorphous microwires offers the potential for significantly increased packing density and, consequently, reduced magnetic losses. The proposed powder-metallurgical processing route based on FAST/SPS enables near-net-shape fabrication while preserving the amorphous or nanocrystalline state of the material.			
Rationale for Joint R&D Project	This proposal aims to initiate a collaborative R&D project under the Korea–Germany International Joint R&D Program. The objective is to develop high-efficiency soft magnetic components for EV power modules and AI data center infrastructure based on amorphous alloy microwires, including glass-coated variants. The approach targets enhanced permeability and reduced high-frequency losses, thereby enabling improved energy efficiency in next-generation power electronic systems.			
Key Words	1. Soft magnetic components 2. High frequency 3. Amorphous wires 4. FAST/SPS 5. Metal-ceramic composites			

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	1. Development and optimization of Fe-based and Co-based amorphous alloy microwires for soft magnetic applications. 2. Development of FAST/SPS-based processing routes for the fabrication of dense soft magnetic components integrating amorphous microwires. 3. Demonstration of prototype soft magnetic components suitable for EV power electronics and AI data center applications, with improved high-frequency performance.			
R&D Target Specifications	A stacking factor ≥ 80% is targeted for amorphous microwire-based soft magnetic cores. Magnetic losses are expected to be significantly reduced, particularly in the high-frequency regime relevant to modern power electronics (kHz to low MHz range). In addition, tuning of the hysteresis behavior (permeability) through processing pressure will be investigated to enable process-based component design.			
R&D Plan	Plan 1: Microwire Production Korea: Produce glass-coated microwires using the Taylor process. Germany: Develop specialized soft magnet component processes. Plan 2: Development of soft magnet composite materials Korea: Provide integrated multi-microwire composite material technologies. Germany: Develop processes for soft magnet component production especially FAST/SPS. Plan 3: Manufacture of soft magnet components and Verification Korea: Manufacture prototypes and perform initial tests. Germany: Develop FAST/SPS-based processes for the fabrication of soft magnetic components and perform detailed magnetic characterization. Equitable technological contributions from both partners (approximately 50% workforce allocation each). Project duration: 36 months (3 years). Key project phases: Microwire production → Wire twisting and braiding → Composite material manufacturing → Comprehensive property evaluation.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IFAM has long-standing expertise in powder metallurgy and advanced coating technologies for functional materials. For more than two decades, the institute has developed and characterized soft magnetic materials and components, covering process development, materials design, and microstructural and functional characterization. Within the joint project, Fraunhofer IFAM will lead the development and manufacturing of soft magnetic components suitable for EV modules and AI data center applications, including comprehensive magnetic characterization and system-level evaluation in accordance with European energy efficiency standards.
Expected R&R of Korean SME Partner	The Korean SME partner is expected to provide advanced processing capabilities for amorphous microwires with different alloy compositions. In particular, advanced production technologies for Fe-based and Co-based amorphous microwires will be addressed. The partner is further expected to contribute expertise in the development of microwire-based structuring methods, including weaving, twisting, and braiding. Comprehensive materials characterization and evaluation capabilities are required to support systematic analysis of composition–process–property relationships.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project is expected to enable market entry into the specialized sector of soft magnetic composite materials by establishing advanced amorphous microwire-based technologies. Through the close collaboration between the German and Korean partners, the joint use of research facilities and testing infrastructure will be strengthened, fostering sustainable international cooperation. The development and refinement of advanced microwire production and processing techniques are anticipated to improve market access in both Asian and European markets by enabling high-performance metallic fiber products with superior magnetic permeability and functional performance compared to state-of-the-art soft magnetic materials. Beyond high-efficiency soft magnetic components, the project is expected to expand the range of potential applications into additional fields of power electronics, energy conversion, and emerging electromagnetic systems. Overall, innovative microwire structuring and integration technologies, such as weaving, twisting, and braiding, are expected to create new opportunities for market penetration and commercialization.

6. References

(1) M. Thamm et al., "Microstructure and magnetic properties of Nanomet compacted by spark plasma sintering", Journal of Magnetism and Magnetic Materials, S. 172121, 2024, doi: 10.1016/j.jmmm.2024.172121.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization